

HACHINOHE, Japan

WMO: 475810

Lat: 40.5275N

Lon: 141.5219E

Elev: 94

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.2	(c) 21.4	(d) 10.5	(e) 9.4	(f) 24.4	(g) 12.4	(h) 10.4	(i) 25.8	(j) 30.9	(k) 33.1	(l) 28.1	(m) 32.2	(n) 11.2	(o) 240	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.2	(c) 87.8	(d) 75.1	(e) 84.4	(f) 73.5	(g) 81.3	(h) 72.3	(i) 76.4	(j) 84.6	(k) 75.2	(l) 82.3	(m) 73.9	(n) 80.0	(o) 14.0	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	126.9	80.1	72.9	122.3	79.3	71.7	117.5	78.5	39.9	84.6	38.8	82.5	37.6	80.1	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 29.5	(b) 25.5	(c) 22.1	(e) 16.0	(f) 93.4	(g) 2.1	(h) 2.6	(i) 14.5	(j) 95.3	(k) 13.2	(l) 96.8	(m) 12.0	(n) 98.3	(o) 10.5	(p) 100.2		
			DB	WB	14.9	78.4	2.1	1.3	13.4	79.4	12.2	80.1	11.0	80.9	9.5	81.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	30.6	31.7	37.6	47.7	56.9	62.5	70.2	73.5	67.5	56.5	45.2	35.3
	DBStd	15.84	4.52	5.19	5.93	6.69	6.56	5.71	6.42	5.66	5.44	5.39	6.45	5.27
	HDD50	2276	603	514	386	122	15	0	0	0	0	8	171	457
	HDD65	5586	1068	934	848	520	262	113	22	3	32	268	594	922
	CDD50	2779	0	0	3	53	229	375	627	730	525	210	26	1
	CDD65	612	0	0	0	1	11	38	184	268	106	4	0	0
	CDH74	3522	0	0	0	10	82	169	1082	1704	468	7	0	0
	CDH80	936	0	0	0	1	13	28	305	494	95	0	0	0
(14) Wind	WSAvg	10.0	12.0	11.9	12.3	11.6	9.7	7.6	7.2	7.6	8.3	9.6	10.9	11.8
(15) Precipitation	PrecAvg	43.5	2.2	1.9	2.4	2.5	3.5	4.1	5.2	5.2	6.5	4.2	2.9	2.4
	PrecMax	56.9	5.4	6.7	4.7	6.4	6.3	9.3	10.9	12.7	12.7	8.8	6.4	6.5
	PrecMin	32.0	0.7	0.6	0.6	0.5	1.1	1.1	0.5	0.7	1.7	0.8	0.6	0.8
	PrecStd	6.2	1.2	1.2	1.1	1.4	1.5	2.2	2.6	3.3	2.8	2.4	1.6	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	52.7	62.9	74.9	81.0	84.1	91.1	92.4	87.6	74.6	66.9	55.5
		MCWB	41.1	44.1	50.6	56.5	60.9	67.4	76.0	76.7	75.0	64.1	56.9	49.3
	2%	DB	42.7	46.7	56.3	67.9	75.8	78.4	87.1	88.7	82.4	70.8	62.2	50.7
		MCWB	37.5	40.1	46.2	53.2	59.9	65.9	74.4	76.0	72.7	61.8	53.9	45.4
	5%	DB	40.0	43.2	51.6	63.0	71.8	75.0	83.3	85.4	79.0	68.3	59.0	47.2
		MCWB	35.4	37.3	43.7	50.5	58.4	64.8	73.2	74.9	70.7	60.5	52.2	42.4
	10%	DB	37.7	40.2	47.9	58.8	67.8	71.7	80.0	82.2	76.1	66.0	56.1	43.9
		MCWB	33.6	35.1	40.9	48.8	57.0	63.6	72.0	73.6	68.9	59.1	50.2	39.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	45.7	51.3	58.5	65.2	71.4	77.2	78.3	76.4	67.7	59.1	50.5
		MCDB	45.3	51.3	60.2	68.9	75.2	78.3	87.4	87.8	84.2	70.9	63.5	54.5
	2%	WB	38.4	40.9	47.4	55.1	62.5	68.5	75.5	76.9	74.5	64.3	55.4	46.2
		MCDB	41.9	46.0	54.9	64.7	70.9	75.4	84.1	85.5	80.5	68.2	60.7	50.1
	5%	WB	36.0	38.0	44.6	52.5	60.2	66.3	74.2	75.9	72.5	62.1	53.1	42.8
		MCDB	39.1	42.4	50.8	60.2	69.0	72.6	81.7	83.3	77.6	66.6	57.7	46.8
	10%	WB	34.1	35.4	41.5	50.1	58.1	64.4	72.8	74.7	70.2	60.2	50.8	39.7
		MCDB	37.0	39.5	47.2	57.8	65.9	70.5	79.2	81.1	74.5	64.7	55.4	43.6
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	12.1	13.8	16.0	15.3	12.5	11.5	11.2	12.9	15.3	14.5	11.5
		MCDBR	13.7	16.9	20.5	22.6	22.6	18.5	17.2	16.2	15.8	17.2	19.1	16.4
		MCWBR	10.4	12.3	13.6	12.4	10.8	8.6	7.4	6.5	8.0	10.3	12.9	12.6
	5% WB	MCDBR	13.5	16.5	19.7	20.1	19.7	16.2	15.8	14.5	14.3	13.8	17.3	16.3
		MCWBR	11.0	12.8	14.2	12.6	11.0	8.6	7.4	6.3	7.8	9.6	13.1	13.4
(40) Clear-Sky Solar Irradiance	taub		0.366	0.410	0.481	0.530	0.545	0.526	0.528	0.500	0.444	0.416	0.397	0.376
	taud		2.066	1.944	1.855	1.825	1.865	1.994	2.062	2.149	2.261	2.211	2.160	2.107
	Ebn at Noon		238	244	240	239	239	244	242	245	252	243	227	223
	Edh at Noon		36	48	59	64	63	56	52	46	39	36	33	33
(44) All-Sky Solar Radiation	RadAvg		510	736	1077	1413	1561	1530	1363	1306	1133	889	591	435
	RadStd		23	60	77	126	162	115	154	162	91	78	43	30

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	+0.85	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page